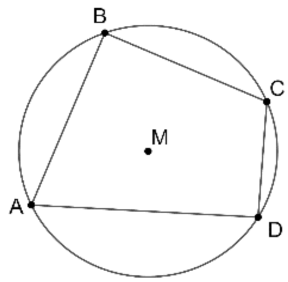
Geometry
Unit 11
Lesson 4 Practice

Name____________________
Date_________________
Period__________

Circle M is shown, with $ABCD$ inscribed in the circle. The $m\angle ABC = 96°$, and $m\angle BCD = 125°$.



1. Find the $m\angle ADC$.

$180 - 96 = 84°$; $m\angle ADC = 84°$

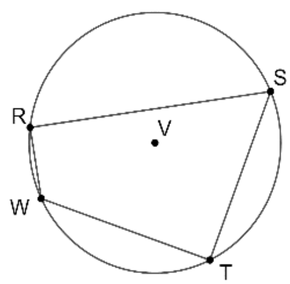
2. Find the $m\angle DAB$.

$180 - 125 = 55°$; $m\angle DAB = 55°$

3. Find the $m\widehat{ABC}$.

$2 \cdot 84 = 168°$; $m\widehat{ABC} = 168°$

Circle V is shown, with $RSTW$ inscribed in the circle. $m\angle RWT = 140°$, $m\angle WRS = (5x)°$ and $m\angle RST = (2x + 6)°$.



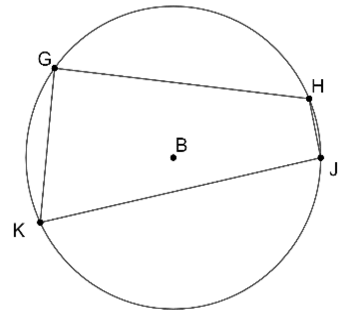
4. Find the value of x.

$(2x + 6) + 140 = 180$
$x = 17$

5. Find the $m\widehat{WTS}$.

$2 \cdot 5(17) = 170°$
$m\widehat{WTS} = 170°$

Circle B is shown, with $GHJK$ inscribed in the circle. $m\angle GHJ = (3x + 50)°$, $m\angle HJK = (4x)°$, and $m\angle GKJ = (4x - 10)°$.



6. Find the value of x.

$(4x - 10) + (3x + 50) = 180$
$x = 20$

7. Find $m\angle KJH$.

$4 \cdot 20 = 80°$; $m\angle KJH = 80°$

8. Find $m\angle GKJ$.

$4 \cdot 20 - 10 = 70°$; $m\angle GKJ = 70°$

9. Find $m\angle HGK$.

$180 - 80 = 100°$; $m\angle HGK = 100°$

10. Find $m\angle JHG$.

$3 \cdot 20 + 50 = 110°$; $m\angle JHG = 110°$